

ActInf GuestStream 081.1 Hannah Biddell, Ruben Laukkonen: Arousal coherence

GuestStream | 59:06

Hello and welcome. This is Active Inference Guest Stream 81.1. It is April 23rd or 24th, depending on where you are. And we're here with Hannah Biddell discussing Feeling Our Way Through Uncertainty. Thank you for joining Hannah. We'll have a presentation followed by a discussion section. So looking forward to anyone's questions in the live chat and thanks again for joining and looking forward to it. Thanks so much, Daniel. I'm really looking forward to it too.

Like I said before, I've been a big fan of the YouTube channel for a long time. So it's really nice to be here. So today I'm going to talk about some of my work in my PhD, which is about arousal coherence, which will make more sense in a moment. But the basic topic is how we might feel our way through uncertainty through the link between affect and interoception. So gives me one minute. All right. So I'm going to be talking about this paper that I wrote with Mark Sohn's, Aline Slagda and Ruben Laucanen, both in Neuroscience of Consciousness recently. And we'll talk about it kind of step by step. I will note at the get-go that there's no formal modeling yet for this concept. So this talk and the paper is basically a series of claims, which I kind of back up with other people's experiments, models and simulations. So I'm just piecing together some prior ideas, basically, and looking at where those might lead. In particular, interceptive self-inference, deep affective inference, and, excuse me, affect in terms of precision optimization.

And if those aren't clear yet, they might be soon, hopefully. So these are the basic claims that I'm going to make. And it's going to be quite a quick talk because there's only five. So firstly, that your autonomic arousal states track with changes and uncertainty, and that means multiple things. And I'll break down some of those in the next slide. That accurately inferring autonomic arousal is important for the way that our brains estimate uncertainty. And that affective arousal might optimize the way that how confident we are in autonomic prediction errors. So it's going to affective arousal, and that affective arousal will also impact potentially the accuracy of our interoceptive autonomic inferences. Which will lead me to the kind of point of the paper, which is arousal coherence. So that is, you can kind of see here, this is a very simple depiction, oversimplified depiction of what arousal coherence would look like. So we have changes in autonomic arousal across time, increases and decreases. Coherent affective arousal would be changing with the autonomic arousal, and discoherent affective arousal would be dissociated from changes in autonomic arousal.

Like, it is not this simple, obviously, um, those layers of depth to change. So the simple time axis is an oversimplification, but this is a basic idea of arousal coherence. And I think and have argued in this paper, that it impacts and reflects our capacity to adapt to changes in uncertainty, and of multiple different kinds. And the final point is meta awareness of arousal. So knowing your arousal, or having a higher order representation of your arousal, might enhance arousal coherence. And I'll talk about some of the benefits associated with higher arousal coherence too. Just quickly to mention the way that I am conceiving affective arousal is a lower order prediction. So it's not the same as emotion,

like stress, anxiety, excitement, but a high arousal is common across those kinds of emotions. Whereas a relaxing state, for instance, would be a low arousal. And if we're talking about autonomic arousal, um, I'd like to clarify that I'm not being very specific here about whether we're talking about sympathetic arousal, sympathetic activation, or parasympathetic, or the balance between those. Just broadly, increases of autonomic arousal or decreases, um, conceived through things like heart rate increase or changes in pupil dilation. Um, okay. So, um, I have not included as yet a, uh, a primer on active inference, given the audience and given the channel. Um, the basic concepts, uh, would be that I'm conceiving of action and perception as inferences, um, in a loop kind of inseparable. Um, also looking at inferences as deep. So, uh, one layer of inference, uh, can be nested within other layers of inference. And that will come up, especially as we start talking about affect. And a third concept from the active inference literature is the idea of precision as, uh, an inverse uncertainty estimate. And so confidence in signals or confidence in beliefs.

Daniela

different, um, lines of evidence that support this and they date back quite a while. Um, the work of, um, you and Diane comes to mind in particular. There was a lot of, uh, work in the early 2000s where they looked at different neuromodulators of autonomic arousal, um, in particular, uh, adrenaline or norepinephrine and changes in pupil dilation that relate to that, that track with changes in uncertainty. Uh, we can also talk about the changes in volatility, changes in surprise, violations of precise expectations. So we see these embodied changes in response to small changes in environment, in uncertainty in the environment. Um, I won't call it environmental uncertainty because it's not necessarily precise. Uh, one of my favorite papers in this area was by the Berker and his colleagues at UCL and they had a task, a probabilistic learning task. Uh, so

[illegible]

we're better able to learn

how well you track uncertainty and how well you can use that information.

Another piece of evidence there is that, I can't remember the chemical's name, Propofenol I think is the name of it, suppressing your autonomic arousal responses, so your pupillary and cardiac arousal reduces learning when the environment changes. So you're you it looks like the effects of prior beliefs increases so you're not updating your uncertainty as well when you don't have these embodied changes and that track with uncertainty. So that's the first claim the autonomic arousal states track with changes and uncertainty and it matters that they do. Second claim is that inferring our autonomic arousal states informs uncertainty estimates.

So here I'm drawing pretty heavily on the work of Mika Allen.

I really enjoy this interceptive self-inference idea which is that by inferring our autonomic arousal states we can better estimate changes in expected uncertainty.

So at the simplest level things like your heartbeat come with changes in sensory noise.

So for example you get pulsatile blood flow across your retina and you can't see in that instance as well.

So by anticipating those changes in your heartbeat the brain can or by interceptively self-inferring them the brain can adjust expected precision accordingly.

So if your heartbeat is going to impact precision then your heart rate over time will also have some impact on uncertainty there by extension.

So there's different levels when we're talking about autonomic states and in terms of the actual state and then we have rhythms and etc.

Another interesting thing to note with this idea of interceptive self-inference is that if the brain can infer these expected states it can likely also control these states.

So using the body to control precision trajectories.

So depending on the contextual need in the environment the body can change depending on uncertainty.

So we saw some indication of this in the previous slide.

These changes where the body is tracking with uncertainty.

Maybe operating through a basic inactive interoceptive self-inference.

Another example is that with perceptual uncertainty your heart rate slows.

So it's a high-precision and which increases the amount of time and predictability of the diastole aspect of your heartbeat.

And therefore high predictability of and opportunities for high precision sensory sampling.

So anyway that's interceptive self-inference.

I think it's a fantastic idea and I want to basically with this just just layer up and look at what might be impacting interceptive self-inference and in particular interceptive precision.

So in order to infer these interceptive conditions we need to be able to adapt interceptive precision when things change.

So typically interceptive precision would be considered quite high.

You have a lot of homeostatic states that you need to stay in.

You need your heartbeat within a certain range in order to survive for example.

But things do change and you need to be able to adapt how confident you are in your interceptive beliefs and interceptive errors.

And this seems to have pretty big implications for health.

So Ryan Smith's work recently on this I found really interesting.

They gave people a heartbeat tapping task.

So just trying to detect their heartbeat during an interceptive perturbation.

So holding their breath.

And their modelling showed that where healthy individuals change their interceptive precision when things change.

People with symptoms of anxiety, depression, substance disorders and eating disorders didn't change their precision estimates from resting levels.

Okay.

So.

That's.

Claim two.

We can use autonomic states to inform our uncertainty estimates.

Claim three.

Claim three.

And this is a.

Based on a kind of gestalt of work in affective inference.

So there's a lot of people.

Working on valence in this area and I've seen some really beautiful work and in particular I really like.

Caspar Hesp et al's paper on this.

I'm considering higher order or affect as.

A deeper or.

Trying to think of the opposite of nested.

An embedding of lower order.

Inferences.

So representing changes in lower level uncertainty at a higher level.

As a kind of abstraction.

And that is what affect.

Is.

We can talk more about the specifics of valence.

After but.

The key that I want to draw attention to here is that.

Deviations in this uncertainty.

A lower level uncertainty or lower level of errors.

But the key thing here for me was that the afferent information from the body to the brain in that coupling, that information flow was modulated by affective arousal in particular, and not valence or anything else.

So how aroused you said you felt was modulating how much information was kind of coming up to your brain from your body after the initial kind of activation.

Okie dokie.

I apologize. This one's a bit text heavy, but we're almost at the end.

So this is where arousal coherence comes in.

Again, the relationship between or correlation between autonomic arousal and affective arousal within a person across time.

So, recapping a little bit, if affective states recruit bodily signals or specifically the outcomes of interceptive inferences, errors or not, to model the current estates of affairs, and changes in autonomic arousal reflect and enact changes in uncertainty and precision.

So, if affective arousal, I'm suggesting, might reflect this minimally metacognitive, to quote Pes, a mechanism through which the precision of interceptive self inference can be regulated and represented at a conscious level.

And I think that this is important and ties back potentially to the work with Ryan Smith and this page here, sort of failure to adapt interoceptive precision in response to autonomic changes.

I think this is where affective arousal comes in and this coherence is the capacity to adapt, interoceptive precision according to changes in autonomic states.
states.

And I think based on the relationship between autonomic arousal and uncertainty, then that arousal coherence might shape the responsiveness, especially at this conscious level, to changing levels of uncertainty.

So, through the body into the environment.

And then if we can monitor and track the changing uncertainty, then we can better use it to shape our learning.

And so we'll see better adaptation to the environment, we'll see better mental wellbeing, et cetera.

And there is some evidence for that.

Although with the caveat that most of the research that's looked at what I'm calling arousal coherence here, we'll pick a autonomic marker.

So heart rate, pupil dilation, something like that.

And then usually it's a emotion that is used to track.

So stress, anxiety, excitement, regardless of valence, some emotion.

So there's a slight caveat there in the methodological domain that we're talking about affective arousal and not emotional.

Emotion exactly.

But there is quite a decent literature now showing that something like arousal coherence is linked to better emotional awareness, better self-regulation, better wellbeing, lower anxiety, lower depression, lower inflammatory biomarkers, et cetera.

So there's some research there pointing towards this being potentially the case.

there are methodological constraints. So I'm expecting and testing whether arousal coherence is associated with more adaptive updating of uncertainty estimates. The paper by DeBerker and colleagues that I spoke about at the very beginning, the probabilistic learning paper, they have kindly shared their data with me. So I've been able to have a look at how coherence might relate to that. I think it's bad form to talk about results before they're written up. So maybe I won't, but I'll dangle that there. There's, there's something with probabilistic learning. I'm also looking at how arousal coherence relates with emotional awareness and emotional self-regulation again. So extending on prior work by people like Sasha Somersfeld and Richie Davis. And they've kind of already found these wellbeing links, but I am wanting to look specifically if it has to do with higher order affective, like updating higher order affective representations and not getting stuck. And then another direction would be that interventions, increasing better awareness of arousal. So mindfulness, hormonal practices, things like ice baths, breath work would lead to increased arousal coherence. Excuse me. And basically that cultivating arousal coherence will help us to navigate uncertainty. That's, that's the key. We can fill our way through it. And just two additional things to note. And one, this data is readily available pretty much everywhere. So there's, there's so many papers out there that have looked at high arousal emotions or just affective arousal in conjunction with some measure of autonomic arousal across time and some simple linear mixed modeling.

So I think you can look at arousal coherence in relationship to various indicators, which is what I've done. My entire PhD is secondary data.

So if you have some of that data and you want to look at it, let me know.

And, uh,

you know, actually, I think I'll leave it there because, uh,

the other things I imagine might come up in the, when we chat anyway, while I have a chance, just another opportunity to say thanks to these legends.

And I was so delightful to get to work with Ruben, Helena, Mark on my first, first author paper. Um, what a treat.

And also two very helpful reviewers who gave much more feedback and much more helpful feedback than I had been led to expect by all the reviewer two memes.

And that's it.

Cool. Thank you.

Um, you could leave the slides up and yeah, let's discuss.

And if anyone has questions, they can ask and I'll read it.

Um, okay.

While people are typing their questions, like what led you to study this intersection of topics?

For your PhD.

Uh, so I was really interested at the very beginning of my PhD.

It's been quite a journey about why some people are better, um, at dealing with uncertainty than others.

I guess you would call it me search.

Um, so I started with looking at when people's worldviews are threatened and, and some people can kind of

roll with that, take that on board and then other people lose their mind.

Um, and I suspected that it had something to do with embodiment based on just looking at the people around me who were better at dealing with uncertainty.

Um, and then I came across, uh, Carl Friston's work and started looking at free energy in particular and, and links between, uh, making links in my mind between free energy and arousal.

Um, and then fortuitously came across Sasha Sommerfeld's work.

So these things all happened within the span of a month and trying to answer this first question and then received the framework and then.

Looked at the measurements.

It's like, ah, these really seemed to add up together.

Um, so it was just really, really nice timing in terms of the way that I was reading things.

Cool.

Um, yeah, I mean, I've just been, I think I've just been really lucky about the time in my, uh, in the intellectual field that I've kind of arrived in.

Because all of these ideas are, um, as you might've seen, just, there's just other people's ideas that I really like.

I've just put like this on top of each other.

Yeah.

Yeah.

Like the novelty and the speed and other factors, the unifying kind of spirit make it so that the layering is very fine and very fun.

Exactly.

Exactly.

Um, okay.

So to kind of like reflect how I was seeing it and how you layered the argument,

I was thinking about drinking coffee before studying and about how that's this kind of general pharmacological intervention.

And on the first claim you had, one of the pieces of experiment was about the pharmacological intervention.

So it's kind of like, it's an obvious way to think about what these interventions do, but those are entering the body.

And then it's interesting that the directionality of learning, it's like the natural variation in learning and the arousal.

It is happening in the same pattern as the intervention would suggest.

So in that, I kind of like saw the, the bringing together of, of experiments, which like in humans or in any species,

you can never always have exactly the experiment.

Um, so that kind of brought together the idea that the learning rate is related to the excitement in the body.

Um, yeah.

And then that gets very embodied when, again, an obvious kind of corollary of a slower heart rate is more

time in between pulses to have high precision measurements.

So there's like a few kind of just very burst past body bases for heart rate and for pharmaceutical interventions that otherwise the connection between like the body's status and even the learning rate can feel very abstract when it's like the least abstract thing possible.

Yeah.

Yeah.

And I think Mika Allen's work on that has been really helpful for me to conceive of that.

Like, okay, obviously your brain needs to predict what's going on in your body so that it can account for those changes that your, your body, the impacts that your body has on uncertainty.

Um, another thing that, that I've just been reminded of is as you were talking about the learning rate is, um, and so I, I'm conceiving of this as like embodied extent uncertainty processing extended into the body.

So predictive processing is not restricted to the brain.

It's a whole, it's an organismic process.

Um, but that said, these things that we're talking about, the different, the changes in autonomic, um, states might be represented also in the, well, likely are represented also in the brain.

So not going to make a claim about necessarily if it's all brain or if it's also brain and body, I think body, but the locus coeruleus, um, so the key source of noradrenaline, the arousal kind of hormone in the brain.

Uh, there was a lovely paper by sales and colleagues.

I think it was 2019 where they showed locus coeruleus, um, activity tracking with Bayesian surprise as well.

Also linking, uh, with learning rates.

So I think arousal, arousal and learning rates are very clearly linked.

Yes.

Okay.

We're back in the coffee and the studying and the coffee.

Just have another sip.

And we're, we're, we're learning through time, hopefully about like how excited or revved up to be for the learning rate of the topic that we're studying.

And then in figure one, the, the precision had this kind of swoop because it went slowly down as the coherence increased, but then at the very end, it kind of ticked up.

So what made you have that little uptick there or what was that representing?

So that I, um, I didn't mean as much by it as you might have done by it though.

I'm sorry.

I was trying to, uh, with these two, these different levels kind of indicate that these are different time scales.

Like, I think it could be a, um, slip that informative.

It could be, it could be a Freudian slip.

On, uh, on the far end of like synchrony.

That's also an uninformative state.

That's very true.

Yeah.

And so it becomes the, the precision explodes.

So it's kind of like there's one impulse is to have higher sensory resolution, but then that is accompanied by, um, lower heart rate.

So lower ability to like do motor activity.

So if you're doing long exercise or something like that, like you, you couldn't necessarily just switch on a dime, but then at the slower timescale, you, you could relax to have better sensory acuity.

But if you're moving fast, it doesn't necessarily matter as much.

But then if you're like too coherent, you're like having like a sleepy seizure or something like that.

Yeah.

I think that this is a, this is a really important point and, and the paper kind of might gloss over it a little bit.

Um, but with the, in general arousal coherence, I think is a positive state, but not necessarily all the time.

So we talk a little bit in there about how, um, in certain kinds of processing, like for example, in like unrelenting grief or really acute stress.

It's probably more adaptive to turn off.

Like to down weight your interoceptive signals there and not be coherent.

And, and we'd see, um, grief and suppression are typically associated with lower coherence.

So it's not that it's always for the best.

Um, I think this like capacity of meta arousal is, is likely more important.

So being able to adjust whether it's coherent or not, but on average coherence is a better state to reside in, uh, or in terms of getting more information or being able to navigate uncertainty better.

Yeah.

But that's not necessarily what, what we always need to do.

Yeah.

Sometimes we just need to be stable and, you know.

Yeah.

Like shock.

It's kind of like some, something's just stuck.

And so the fact that the mutual information between the body's state and the cognitive state, the mutual information being low, it's not necessarily a bad thing.

It's like a mechanism.

Protective.

Yeah.

It's a protective and it, and it can be processed later potentially, but that's.

Um, then kind of moving through the claims, getting to where, um, we have the, the consciously representable nature of arousal.

Um, and it isn't arousal itself that we directly perceive.

This is kind of thinking about the interoception as inference because we could have kind of a first pass, like how excited am I?

But then check in and like find out otherwise, or have the biomarkers otherwise, like trying to guess your, your blood pressure.

So then I thought about studying with a coffee and a lot of the things with the pupil, which represents the arousal.

And then another eye related phenotype with the saccade, which represents the epistemic value, those movements and, and muscle contractions are basically subliminal.

Most people, probably almost all people cannot control their.

Yeah.

Yeah.

So there's a layer that is slower than reading across the page that we could access.

And that's that kind of like arousal layer.

So it just is like, there's things that are too fast.

Like if it were otherwise, then we'd be having a different live stream asking why we could perceive something else.

But in the ways that we have our attention now, there are things that are just happening faster that we don't pay attention to like syntax and letter recognition.

And then this is kind of moving into, okay, well, what is that slower process that is necessarily slower and different than the primary kind of like domain specific inference?

I mean, I'm thinking of that as affect, I guess, as one instance.

Um, I think there's also an argument to be made that attention is doing a similar job.

Uh, yeah.

Have I understood your question correctly?

Yeah.

Yeah.

Yeah.

Yeah.

We're just trying to make sense.

So if you're thinking about like the lowest level, for example, being heart beat and then heart rate and then, uh, changes in heart rate, according to.

Something at a bigger level.

I don't know, some other cycle, circadian rhythm.

Yeah.

That's what I was going to ask.

What is the arousal and affect nested within?

And what does that mean?

Uh, arousal as in autonomic or arousal as in affective nested within?

Cause I, I, I treat them as different.

Okay.

Maybe clarify it then.

Uh, how do you see it?

Yeah.

Sorry.

I'm thinking of autonomic arousal as a lower level inference than affective, but higher level than for instance, the actual autonomic, uh, interoceptive inferences.

So you're, uh, you're inferring a heartbeat for instance, or when your heart expecting that, and then above that expecting things like, um, heart rate or changes across time.

Uh, and then up at higher levels and it's not a precise, this level, this level.

Um, but up above that affective inference as a kind of minimal form of better cognition, including over interoceptive information, but not exclusive to interoceptive information.

Um, so you can have affective states, even if your interoceptive changes or like autonomic changes are suppressed.

Hmm.

Hmm.

And then, uh, above affective states, I'm imagining the, the metacognition kind of the act, not minimal, but actual meta awareness of those around affective arousal states.

Like knowing that we know those states or more like being able to actually, well, both, uh, being able to actually report on those states, I think is very different from feeling those states.

And the subjective feeling experience, which, um, some is talking about as like your minimal consciousness.

The feeling of being is a very different thing from being able to say what you feel.

And self report on that.

In my opinion, there's some other people who agree with me.

There's probably people who don't.

That's fine.

Uh, so, um, yeah, a bit of nesting, nesting, nesting.

That's how I'm conceiving of this.

And it's not, uh, by any means because the modeling hasn't been done.

I think it's not a precise, this was layer one.

This was layer two, this layer three.

Um, by just pointing that these things do appear to be nested within each other.

Hmm.

And of course have a bi-directional relationship.

Yeah.

It's very interesting.

Um, a big motivator or I guess kind of pragmatic implication.

You said that having a better representation that was conscious, whether just naturally like variation in how well people can self represent, um, or the trainings that, that might be possible.

Like it could be used to shape our responsivity to the environment and learning.

So like, what would it look like or what would that mean?

So I think that's the best way to answer that question.

I think, uh, the simplest way to put it is that we know the world through our body.

Like our body is our medium of relating to the world.

And so interoceptive influences are necessarily kind of involved with how we perceive what's going on outside of us.

Um, by being kind of, we think about coherence as a kind of interoceptive accuracy of changing conditions.

So you're integrating, it's like an integrative accuracy over interoceptive conditions or autonomic conditions.

That accuracy shapes, for example, interoceptive self inference and the capacity of the brain to predict, you know, the pulsatile blood flow when that's going to happen over your eyes.

So impact how it adjusts precision according to changes that are actually happening in your body and how well it can enact those changes.

according to the needs of the environment.

And by that, I mean changes in uncertainty in the environment.

Does that make sense?

I'm not sure if I rambled a little bit there.

Like, I think it's a very, it's a very exciting bundle.

It's a very exciting bundle because these are things, these are cycles that we go through on minute by minute and day after day.

And that where that kind of took me and what reading the paper made me wonder about was like entering in that learning example, studying for the, from the textbook, but also that could be, um, a squirrel learning about the environment, or it could be having a memory of a new experience.

So it's not just like a domain learning, but it's like, you could kind of focus attention on the, the primary inference problem.

Like look harder at the letters.

So there's a kind of like attention to the primary sensory, um, aspects, which on one pass, it's kind of like, well, that's how you interpret the word better.

But this is like suggesting that there's also like a higher order entry point for improving our learning, which would be like to be monitoring our own awareness and, and our embodiment.

And like, well, that equation made me, made my heart rate increase just to see it.

Or this is making me sleepy.

I'm bored.

This is making me frustrated.

Like those kinds of inferences, they're not a waste of time.

Um, because they would actually set up the scaffold for the more adaptive interpretation of the lower level sensory data.

But if you just entered in at the sensory data, you might be missing out on the affective part.

Yes, I agree.

I think that's a, I think it's a really exciting area for like looking into in the future, because we can, we can test

these.

things, you know, we have that capability to test people's intercepted inferences, for instance, um, and. readily available on mobile.

Um, I'm personally quite excited to go down that.

Oh, um, and I, I agree that.

Um, there's a, yeah, there's something to be.

Gained from treating feelings as information across multiple levels.

And it's not a new idea, obviously.

So, um, there is always this, I mean, there's been this idea, I think that the Schwartz maybe, uh, feelings.

Um, chlorin Steinberg, apologies.

Um, like I said earlier, I didn't have Google, but, um, yeah, feeling, treating feelings as information or seeing feelings as information.

I think is really, uh, valuable and, and not yet explored as much as it could be.

Yeah, not, not, we haven't, we haven't optimized that, I guess.

And I, I think a few ways in which the syntheses that you have kind of.

Though, I don't know all the details of what's been done can maybe go beyond just saying that there's a correlation.

Um, like the two things that come to mind is first off, it goes beyond saying that it, that emotions are affect or awareness are informative because it's not just like, well, you could tune into the, that stream or you could tune into the sensory stream by making them explicitly nested mechanistically.

Then you can, you're, you can learn from both bi-directionally, like, and then a second important piece is, um, by connecting it to the biometrics, like the heart rate and the variability and the pupil, then there's a kind of less emotional entry point.

Like webcams could report on our pupil activity and our sarcades and who knows who already is tracking that information.

But then that gives, it gives an entry point that isn't like asking how one feels, which is a very challenging and involved question to answer and, and, and interocept around.

And some people are terrible at it.

They don't know.

It's, there's big individual variability.

And, and I guess the point of my work is like that variability matters.

It's informative.

Um, how, how well you know, how you feel or how well your, how you feel lines up with your body, I guess is a better way to say that.

Yeah.

And that, that variability, like within and across levels, it can't be, and the goal is not to like clamp it.

Um, because we're always rotating positions, standing up, sitting down, doing all these things that you can't lock on to a value for.

Which is why the kind of skeleton of the model is just about how the variabilities connect rather than like a

number or like a final kind of just.

Yeah.

Equation to describe it across settings.

Yeah.

I would like to come up with some equation.

There will be a nice, uh, a future direction.

I think, uh, I do think that this kind of thing is, um, possibly usable or.

For treatment for assessing the effectiveness of interventions, for instance, um, seeing it like.

Basically, if you can tune into your body more through your feelings.

Whether that's valuable.

Um, I would really like to talk about the, um, experimental evidence that I think would.

Uh, back up what I'm talking about more with the learning external uncertainty.

Um, I like, yeah, maybe I could come back and talk about that when.

81.2.

At least that a pre.

Well, we've got a pre print.

I, I feel like, uh, it's maybe bad form to talk about it in advance.

No, it's, but it's cool to see like how.

Over these even slower, slower timelines.

That the research threads can come together and how they can be brought together.

With a lot of the other.

Technical work that, that is out there.

That is out there.

And then I think that's sort of the question that, that it's exciting to ask is like what data sets are already out there where descriptive statistics can be done.

What generative models can be made.

And just by what you wrote to separate things out into what's an observation and what's a statistical relationship, like the information is all there.

Yeah.

And there's a whole lot more of it.

I mean, reviewers and collaborators are like, please just stop talking, stop talking.

You're saying too much.

Um, so there, yeah, there's a lot there.

There was so much literature that kind of fits into this, um, framework.

Framework out there.

And I think is.

Yeah, it's exciting.

It's exciting and it's easily measurable.

Um, again, with the caveat of the emotions versus, you know, and, and we talked about kind of, you know, self reporting on arousal is not necessarily the same as feeling it.

So really trying to work out whether this difference is happening at the level of self report or happening at the level of feeling or in between that and the interception exactly where that's coming.

It's a little bit difficult to pinpoint because measuring feelings, for instance, is quite difficult.

Measuring self report.

That may be a different thing.

Um, but that said there's, yeah, I I've managed to, I had, uh, three research papers, all preexisting data, all really interesting topics and wonderful people who were just willing to share and collaborate on these.

So, um, looks like this arousal concept relates with, like said, um, well being things and that's reasonably well established already.

But the, I guess the key point, um, that I'm going after is the relationship with navigating uncertainty in the external world, how that, how this relates to that through, through interceptive self inference is my affected interceptive self inference is my, um, my proposal.

Is there anything else you want to add or ask or like mention?

Um, I am very open to talking about it.

I haven't done the modeling.

I don't have a mathematics background.

So, uh, if anyone that does wants to talk about it, please hit me up.

If you have data sets, if you have hypotheses, you want to talk about it.

If you want to see my code, it's on OSF.

Um, yeah, that's it.

Just, yeah, I'm, I'm very keen to talk about it.

Um, see if we can, if I can take it somewhere more interesting, make it a little bit more precise, make it usable.

Thanks so much for having me.

Such a pleasure.

Congratulations on the PhD and.

Thanks.

I haven't got it yet.

Almost, almost, um, until next time.

Until next time.

Thanks, Hannah.

Thanks, Daniel.

[illegible]

Thank you.